

Laboratory for Practicum Students and Commercial

VISION, MISSION & OBJECTIVES

Geomechanics and mine equipment Laboratory are committed in providing high standard services to public with the quality policy implemented by entire staff of the laboratory.

Commitments are expressed in the quality policy as the following:

1. Provide professional testing services with aim of satisfying the customer requirements and in accordance with the standards quality;
2. Entire laboratory staff understood and consistently implement the laboratory management system in accordance with ISO/IEC 17025: 2008;
3. Sustainably improving the laboratory management system effectiveness.

To achieve our commitments, We shall:

1. Establish and maintain continual Occupational Safety and Health Management System (OSHMS) including the relevant resources;
2. Design workplace and the job comply with government legal, regulation and other requirements related to OSH;
3. Provide OSH training and education and awareness to all employees to improve Laboratory's OSH Performance.

CONTACT US

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HISTORICAL DEVELOPMENT OF LABORATORY

In 1996, Head of Mining Engineering Department, Prof. Irwandy Arif assigned Dr. Suseno Kramadibrata to develop a laboratory for practical purpose in Mining Engineering Department ITB. At that period of time, the laboratory facilities was extremely short of equipments. Therefore, Blasting, Drilling and Excavation, along with Mine Ventilation Technique Course held modestly. The first phase of development is laboratory improvement, continued with procurement of instrumentation and equipment by designing and finding used equipments in various mine sites in Indonesia.

In 1997, Prof. Made Astawa Rai as the Rock Mechanics Laboratory Manager asked Dr. Suseno Kramadibrata to do specific repair and improvement of laboratory facilities so that both students and lecturers could do various research in Geomechanics. Rock Mechanics Laboratory support following various courses :

1. Rock Mechanics
2. Drilling and Excavation Technique
3. Blasting Technique
4. Mine Equipments
5. Mine Ventilation
6. Underground Mine Support and Monitoring System
7. Mine Surveying

On 28th of March 2007, Dr. Suseno Kramadibrata received his assignation as manager of the Laboratory of Geomechanics and Mine Equipments, Faculty of Mining and Petroleum Engineering ITB which are divided into 4 (four) departments :

1. Working Group of Rock Mechanics, coordinated by Dr. Ridho Kresna Wattimena
2. Working Group of Mine Explosives & Blasting Technique, coordinated by Dr. Ganda Marihot Simangunsong
3. Working Group of Drilling and Excavation Technique, coordinated by Dr. Nuhindro Priagung Widodo
4. Working Group of Surface and Underground Mine Equipments, coordinated by Dr. Budi Sulistianto

In early 2013, official certification process of the Laboratory of Geomechanics and Mine Equipments from the Ministry of Research of the Indonesian Republic achieved on 11th of June 2013. Certification achieved based on reference from Decree of the Head of the National Standard Accreditation Department No 1976/KAN-HK.37/07/2000 in regard with Establishment of the Indonesian's National Standard 19-17025-2000 as requirements of laboratory accreditation.

In early 2014, Dr. Ridho Kresna Wattimena achieved his assignation as the manager of the Geomechanics and Mine Equipments Laboratory, Faculty of Mining and Petroleum Engineering ITB. This assignation last until 2015 because he had to take place as Vice Dean of Academic Affairs. Afterward, the assignation goes to Dr. Ganda Marihot Simangunsong as the new manager of the laboratory up to present.



LABORATORY OF GEOMECHANICS AND MINING EQUIPMENT INSTITUTE TECHNOLOGY OF BANDUNG



Prof. Made Astawa R.



Dr. Suseno Kramadibrata



Prof. Budi Sulistianto



Prof. Ridho Kresna W.



Dr. Eng. Ganda M. S.



Dr. Eng. Nuhindro P. W.

TESTING

1. Physical Properties

Standard Method : ISRM Suggested Methods For Determining Water Content, Porosity, Density, Absorption and Related Properties.

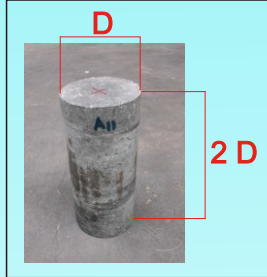
Output : ρ_{nat} (Natural Density), W (Natural Water Content), ρ_{dry} (Dry Density), S_r (Degree of Saturation), ρ_{sat} (Saturated Density), n (Porosity), e (Void Ratio).



2. Ultrasonic Velocity

Standard Method : ISRM Suggested Methods For Determining Sound Velocity (First Method - High Frequency Ultrasonic Pulse).

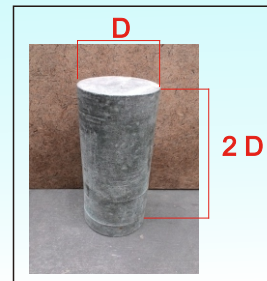
Output : V_p (Compression Wave Velocity), V_s (Shear Wave Velocity), E_{din} (Modulus Elasticity Dynamic).



3. Uniaxial Compressive Strength

Standard Method : ISRM Suggested Methods For Determining the Uniaxial Compressive Strength and Deformability of Rock Materials.

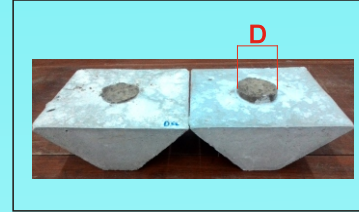
Output : σ_c (Uniaxial Compressive Strength), E (Modulus Elasticity), ν (Poisson's Ratio).



4. Direct Shear

Standard Method : ISRM Suggested Methods for Determining Shear Strength.

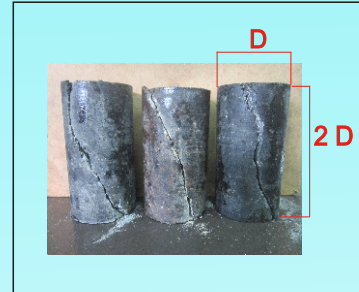
Output : C_{peak} (Cohesion), ϕ_{peak} (Internal Friction Angle), C_{app} (Apparent Cohesion), ϕ_{app} (Apparent Internal Friction Angle).



5. Triaxial

Standard Method : ISRM Suggested Methods for Determining the Strength of Rock Materials in Triaxial Compression.

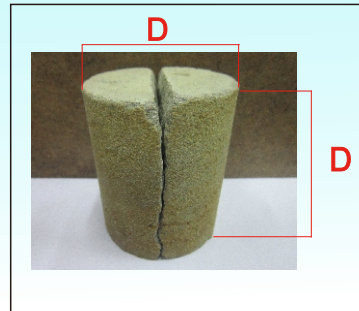
Output : c (Cohesion), ϕ (Internal Friction Angle).



6. Wedge

Standard Method : O & K Wedge.

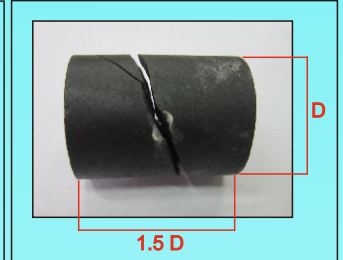
Output : F_L (Digging Force per Unit Length), F_A (Digging Force per Unit Area).



7. Point Load Index

Standard Method : ISRM Suggested Methods for Determining Point Load Strength.

Output : I_s (Point Load Index).



8. Slake Durability

Standard Method : ASTM (D4644-87).

Output : I_D (Slake-Durability Index).



9. Brazillian

Standard Method : ISRM Suggested Methods for Determining Tensile Strength of Rock Materials.

Output : σ_T (Tensile Strength).

